Expense Tracker App Summary

What it is

A Spring Boot and Vaadin expense tracker that lets users upload receipts/invoices, extract data via OCR/AI, and store expenses.

Who it’s for

Individuals or small teams who track expenses and want receipt-assisted entry in a simple web UI.

What it does

- Upload receipt images/PDFs and preview them in the UI.
- Extract text via OCR (Tesseract/OpenCV) or a local OCR API.
- Parse OCR into structured expense fields with AI (Gemini or LM Studio).
- Show expenses with totals and a grid dashboard.
- Persist expenses/users/invoices in MySQL via JPA repositories.
- Expose /api/invoices/upload for raw file OCR extraction.

How it works

Vaadin UI (HomeView, UploadView) -> Spring services (ExpenseService, UserService, OCR/AI services) -> JPA repositories -> MySQL (expenses, users, invoices).
OCR pipeline: UploadView -> LocalAiExtractionService -> local OCR API (localhost:5050) OR OcrService (Tesseract + OpenCV) -> AI parser (Gemini or LM Studio at localhost:1234) -> Expense + Validation.

How to run

- Java 21 is required (pom.xml).
- Configure MySQL at jdbc:mysql://localhost:3306/expense
- with user root and the password in application.properties.
- Start the app: ./mvnw spring-boot:run

Not found in repo

DB schema/migrations; setup steps for the OCR API (port 5050) and LM Studio (port 1234).